

Data Destruction

Sub-techniques (1)

Adversaries may destroy data and files on specific systems or in large numbers on a network to interrupt availability to systems, services, and network resources. Data destruction is likely to render stored data irrecoverable by forensic techniques through overwriting files or data on local and remote drives.^{[1][2][3][4][5][6]} Common operating system file deletion commands such as `del` and `rm` often only remove pointers to files without wiping the contents of the files themselves, making the files recoverable by proper forensic methodology. This behavior is distinct from [Disk Content Wipe](#) and [Disk Structure Wipe](#) because individual files are destroyed rather than sections of a storage disk or the disk's logical structure.

Adversaries may attempt to overwrite files and directories with randomly generated data to make it irrecoverable.^{[4][5]} In some cases politically oriented image files have been used to overwrite data.^{[2][3][4]}

To maximize impact on the target organization in operations where network-wide availability interruption is the goal, malware designed for destroying data may have worm-like features to propagate across a network by leveraging additional techniques like [Valid Accounts](#), [OS Credential Dumping](#), and [SMB/Windows Admin Shares](#).^{[1][2][3][4][6]}

In cloud environments, adversaries may leverage access to delete cloud storage objects, machine images, database instances, and other infrastructure crucial to operations to damage an organization or their customers.^{[7][8]} Similarly, they may delete virtual machines from on-prem virtualized environments.

ID: T1485

Sub-techniques: [T1485.001](#)

❶

Tactic: [Impact](#)

❷

Platforms: Containers, ESXi, IaaS, Linux, Windows, macOS

❸

Impact Type: Availability

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[Version Permalink](#)

Procedure Examples

ID	Name	Description
C0034	2022 Ukraine Electric Power Attack	During the 2022 Ukraine Electric Power Attack , Sandworm Team deployed CaddyWiper on the victim's IT environment systems to wipe files related to the OT capabilities, along with mapped drives, and physical drive partitions. ^[9]
C0063	2025 Poland Wiper Attacks	During the 2025 Poland Wiper Attacks , the adversaries utilized wiper malware to overwrite files using a 16-byte buffer that fully overwrites files 16 bytes or smaller or partially overwrites files greater than 16 bytes to speed up the process. ^{[10][11]}
S1167	AcidPour	AcidPour can perform an in-depth wipe of victim filesystems and attached storage devices through either data overwrite or calling various IOCTLs to erase them, similar to AcidRain . ^[12]
S1125	AcidRain	AcidRain performs an in-depth wipe of the target filesystem and various attached storage devices through either a data overwrite or calling various IOCTLs to erase it. ^[13]
S1133	Apostle	Apostle initially masqueraded as ransomware but actual functionality is a data destruction tool, supported by an internal name linked to an early version, <code>wiper-action</code> . Apostle writes random data to original files after an encrypted copy is created, along with resizing the original file to zero and changing time property metadata before finally deleting the original file. ^[14]
G0082	APT38	APT38 has used a custom secure delete function to make deleted files unrecoverable. ^[15]
S0089	BlackEnergy	BlackEnergy 2 contains a "Destroy" plug-in that destroys data stored on victim hard drives by overwriting file contents. ^{[16][17]}
S0693	CaddyWiper	CaddyWiper can work alphabetically through drives on a compromised system to take ownership of and overwrite all files. ^{[18][19]}
S9042	CanisterWorm	CanisterWorm has targeted wiper components that can delete entire clusters and execute recursive file deletions on non-containerized hosts. ^{[20][21]}
S1134	DEADWOOD	DEADWOOD overwrites files on victim systems with random data to effectively destroy them. ^[14]
S0659	Diavol	Diavol can delete specified files from a targeted system. ^[22]
S9038	DynoWiper	DynoWiper has overwritten files with 16-byte sequences of random data generated by the Mersenne Twister algorithm using the Microsoft Windows native <code>CreateFileW()</code> function to open the file and the <code>SetFilePointerEx()</code> and <code>WriteFile()</code> functions to overwrite the file. ^[23] Additionally, versions of DynoWiper can also delete files using the <code>DeleteFileW</code> API. ^[11]
S0697	HermeticWiper	HermeticWiper can recursively wipe folders and files in <code>Windows</code> , <code>Program Files</code> , <code>Program Files(x86)</code> , <code>PerfLogs</code> , <code>Boot</code> , <code>System</code> , <code>Volume Information</code> , and <code>AppData</code> folders using <code>FSCTL_MOVE_FILE</code> . HermeticWiper can also overwrite symbolic links and big files in <code>My Documents</code> and on the Desktop with random bytes. ^[24]
S0604	Industroyer	Industroyer 's data wiper module clears registry keys and overwrites both ICS configuration and Windows files. ^[25]
S0265	Kazuar	Kazuar can overwrite files with random data before deleting them. ^[26]
S0607	KillDisk	KillDisk deletes system files to make the OS unbootable. KillDisk also targets and deletes files with 35 different file extensions. ^[27]

ID	Name	Description
S0688	Meteor	Meteor can fill a victim's files and directories with zero-bytes in replacement of real content before deleting them. ^[31]
S9043	Mini Shai-Hulud	Mini Shai-Hulud has wiped data on devices that fall within specified parameters to include those that resolve to specific geolocations including Iran and Israel. ^{[32][33]} Mini Shai-Hulud has also implemented a dead-man's switch that wipes the victims home directory if the operator revokes a GitHub token created by the adversary. ^[34]
S1135	MultiLayer Wiper	MultiLayer Wiper deletes files on network drives, but corrupts and overwrites with random data files stored locally. ^[35]
S0365	Olympic Destroyer	Olympic Destroyer overwrites files locally and on remote shares. ^{[6][36]}
S0139	PowerDuke	PowerDuke has a command to write random data across a file and delete it. ^[37]
S0238	Proxysvc	Proxysvc can overwrite files indicated by the attacker before deleting them. ^[38]
S0364	RawDisk	RawDisk was used in Shamoon to write to protected system locations such as the MBR and disk partitions in an effort to destroy data. ^{[3][5]}
S0496	REvil	REvil has the capability to destroy files and folders. ^{[39][40][41][41][42][43][44]}
S9030	SameCoin	SameCoin can overwrite designated files on targeted systems with random bytes. ^[45]
G0034	Sandworm Team	Sandworm Team has used CaddyWiper , SDelete , and the BlackEnergy KillDisk component to overwrite files on victim systems. ^{[46][47][9]} Additionally, Sandworm Team has used the JUNKMAIL tool to overwrite files with null bytes. ^[48]
S0195	SDelete	SDelete deletes data in a way that makes it unrecoverable. ^[49]
S9008	Shai-Hulud	Shai-Hulud has destroyed the victim's home directory by overwriting and deleting every writable file within the user's home folder. ^{[50][51]} Shai-Hulud has also utilized the <code>shred</code> command on Linux devices. ^[52]
S0140	Shamoon	Shamoon attempts to overwrite operating system files and disk structures with image files. ^{[1][2][3]} In a later variant, randomly generated data was used for data overwrites. ^{[5][53]}
G1057	ShinyHunters	ShinyHunters has executed the <code>DeleteBucket</code> API call to delete buckets. ^[54]
S1178	ShrinkLocker	ShrinkLocker can initiate a destructive payload depending on the operating system check through resizing and reformatting portions of the victim machine's disk, leading to system instability and potential data corruption. ^[55]
S0380	StoneDrill	StoneDrill has a disk wiper module that targets files other than those in the Windows directory. ^[4]
G1053	Storm-0501	Storm-0501 has destroyed data and backup files. ^[56]
G1056	TeamPCP	TeamPCP has deployed privileged DaemonSets to delete files on Kubernetes clusters and has executed recursive file deletions on non-containerized hosts. ^[20]
G1055	VOID MANTICORE	VOID MANTICORE has conducted data wiping attacks on compromised systems. ^{[57][58][59][60]} VOID MANTICORE has also manually deleted files from compromised hosts, to include selecting all files and then deleting them. ^{[57][59]}
S0689	WhisperGate	WhisperGate can corrupt files by overwriting the first 1 MB with <code>0xcc</code> and appending random extensions. ^{[61][62][63][64][65][66]}
S0341	Xcheck	Xcheck has destroyed linux-based databases as part of its reconnaissance activities. ^[67]

Mitigations

ID	Mitigation	Description
M1053	Data Backup	Consider implementing IT disaster recovery plans that contain procedures for taking regular data backups that can be used to restore organizational data. ^[68] Ensure backups are stored off system and protected from common methods adversaries may use to gain access and destroy the backups to prevent recovery.
M1032	Multi-factor Authentication	Implement multi-factor authentication (MFA) delete for cloud storage resources, such as AWS S3 buckets, to prevent unauthorized deletion of critical data and infrastructure. MFA delete requires additional authentication steps, making it significantly more difficult for adversaries to destroy data without proper credentials. This additional security layer helps protect against the impact of data destruction in cloud environments by ensuring that only authenticated actions can irreversibly delete storage or machine images.
M1018	User Account Management	In cloud environments, limit permissions to modify cloud bucket lifecycle policies (e.g., <code>PutLifecycleConfiguration</code> in AWS) to only those accounts that require it. In AWS environments, consider using Service Control policies to limit the use of the <code>PutBucketLifecycle</code> API call.

Detection Strategy

ID	Name	Analytic ID	Analytic Description
DET0146	Detection of Data Destruction Across Platforms via Mass Overwrite and Deletion Patterns	AN0411	Adversary spawns command-line tools (e.g., <code>del</code> , <code>cipher /w</code> , <code>SDelete</code>) or scripts to recursively delete or overwrite user/system files. This may be correlated with abnormal file IO activity, registry writes, or tampering in critical system directories.
		AN0412	Massive recursive deletions or overwrites via <code>rm -rf</code> , <code>shred</code> , <code>dd</code> , or wiper binaries. May include unlink syscalls, deletion of known config/data paths, or sequential overwrite patterns.
		AN0413	Destruction via <code>rm -rf</code> , overwrite with <code>dd</code> or <code>srm</code> , often executed by script in <code>/tmp</code> or <code>/private/tmp</code> , may also involve file overwrite to political or decoy image data.
		AN0414	Adversary deletes critical infrastructure: EC2 instances, S3 buckets, snapshots, or volumes using elevated IAM credentials. Frequently includes batch API calls with <code>Delete*</code> or <code>TerminateInstances</code> .
		AN0415	Adversary destroys virtual disks (VMDK), images, or VMs by invoking <code>vim-cmd</code> , deleting datastore contents, or purging snapshots.
		AN0416	Container process executes destructive file operations inside volume mounts or host paths. Includes <code>rm -rf /mnt/volumes/</code> , container breakout followed by host deletion attempts.

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